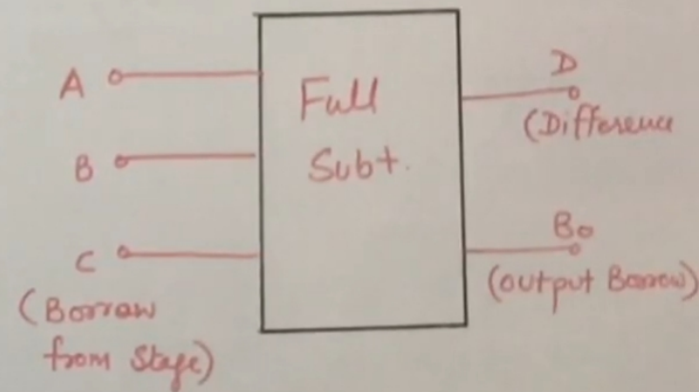


$$2^n = 2^3 = 8$$

FULL SUBTRACTOR

A	B	C	D	B _o
0	- 0 = 0	- 0	0	0
0	- 0 = 0	- 1	1	1
0	- 1 = 1	- 0	1	1
0	- 1 = 1	- 1	0	1
1	- 0 = 1	- 0	1	0
1	- 0 = 1	- 1	0	0
1	- 1 = 0	- 0	0	0
1	- 1 = 0	- 1	1	1

$$\begin{array}{r} 0 \rightarrow 2 \\ + 0 \\ - 01 \\ \hline 01 \end{array}$$



$$2^n = 2^3 = 8$$

FULL SUBTRACTOR

A B C | D B₀

0 - 0 = 0

0 - 0 = 0

0 - 1 = 1

0 - 1 = 1

1 - 0 = 1

1 - 0 = 1

1 - 1 = 0

1 - 1 = 0

K-Map for D

C \ AB	00	01	11	10
	0	1	0	1
0	0	1	0	1
1	1	0	1	0

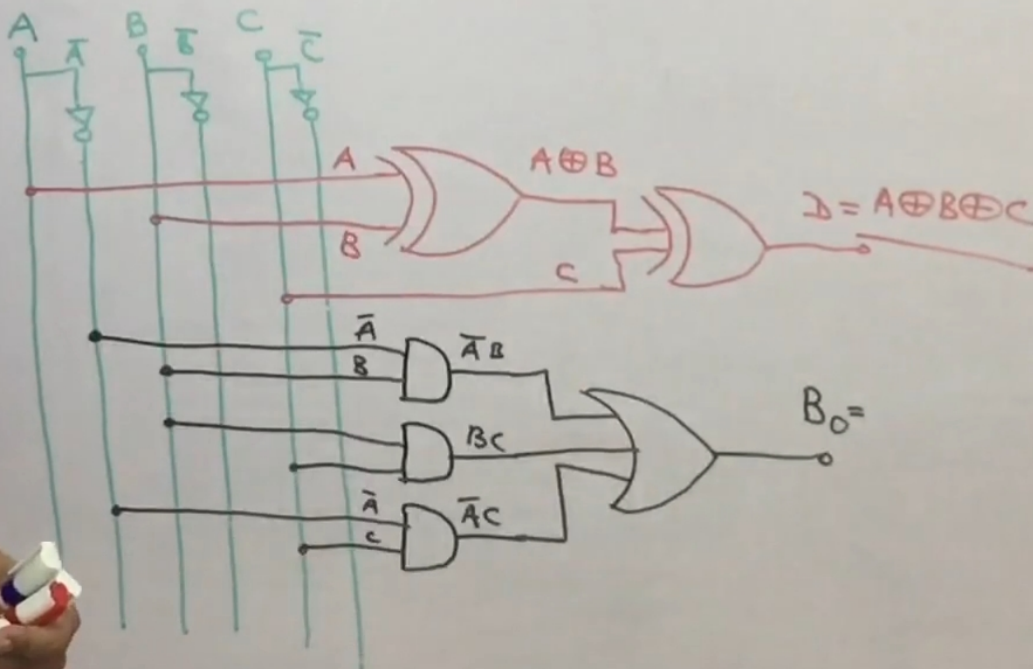
Check Board Conf.

$$D = A \oplus B \oplus C$$

K-Map for B₀

C \ AB	00	01	11	10
	0	1	0	0
0	0	1	0	0
1	1	1	1	0

$$B_0 = \bar{A}B + BC + \bar{A}C$$



FULL SUBTRACTOR

B₀

k-Map for D

AB \ C		00	01	11	10
		0	1	0	1
0	0	0	1	0	1
1	0	1	0	1	0

Check Board conf.

$$D = A \oplus B \oplus C$$

$$D = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C}$$

$$= \bar{C}(\bar{A}B + A\bar{B}) + C(\bar{A}\bar{B} + AB)$$

$$= \bar{C}(\underbrace{\bar{A}B + A\bar{B}}_{\text{Ex-OR}}) + C(\underbrace{\bar{A}\bar{B} + AB}_{\text{Ex-NOR}})$$

$$= \bar{C}(A \oplus B) + C(\overline{A \oplus B}) \quad [\text{Let } A \oplus B = X]$$

$$= \bar{C} \cdot X + C \cdot \bar{X} = X \oplus C$$

$$D = A \oplus B \oplus C$$